

Wind Farm Layout Optimization

Using Aeroacoustic Surrogate Modeling and OpenFAST Validation

Soumyadeep Das
Lehigh University

August 22, 2026

Mentor: Kevin Wyckhoff
Dr. Arindam Banerjee

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Research Objective	1
2	Background	1
2.1	Wind Farm Layout Optimization	1
2.2	OpenFAST and FAST.Farm	2
2.3	FLORIS	2
2.4	NREL 5-MW Reference Wind Turbine	3
2.5	Aeroacoustic Modeling	3
2.6	Decibels and A-Weighting	4
2.7	Surrogate Modeling	5
2.8	Genetic Algorithms	6
3	Methodology	7
3.1	Overall Workflow	7
3.2	Aeroacoustic Surrogate Model	7
3.2.1	Preliminary Surrogate Designs	9
3.2.2	Aeroacoustic Training Data	10
3.2.3	Candidate Feature Representations	11
3.2.4	Model Training and Selection	13
3.2.5	Final Surrogate Implementation	15
3.2.6	Aeroacoustic Aggregation	15
3.3	Genetic Algorithm	15
3.3.1	Chromosome Representation and Initialization	15
3.3.2	Wind Scenarios	16
3.3.3	Fitness Function	16
3.3.4	Selection and Genetic Operators	19
3.4	OpenFAST Validation	21
3.4.1	Wind-Direction Transformation	22
3.4.2	Aeroacoustic Post-Processing	22
3.4.3	Fitness Recalculation	22
4	Results	23
4.1	Surrogate Model Performance	23
4.1.1	Extra Plots	24
4.2	Genetic Algorithm Performance	24
4.3	Optimized Layout	28
4.4	OpenFAST Validation	31
5	Discussion	31
6	Limitations	32
7	Future Work	33
8	Conclusion	33

A Optimization Parameters	33
B Additional Results	33

1 Introduction

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

1.1 Motivation

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

1.2 Research Objective

The main objective of this study is to develop and evaluate a computational framework for wind-farm layout optimization that considers not just power production, but also aeroacoustic constraints. The framework is designed to use computationally efficient models during optimization while also including a higher-fidelity OpenFAST/FAST.Farm simulation for final validation.

More specifically, this study aims to:

1. develop a machine-learning surrogate capable of approximating single-turbine OpenFAST aeroacoustics predictions;
2. integrate the aeroacoustic surrogate with FLORIS and a genetic algorithm to optimize turbine placement across multiple wind conditions;
3. account for turbine-spacing, farm-boundary, and aeroacoustic constraints during layout optimization; and
4. validate the resulting optimized layout using FAST.Farm and OpenFAST, including both farm power production and aeroacoustic performance.

2 Background

2.1 Wind Farm Layout Optimization

Wind-farm layout optimization (WFLO) concerns the placement of turbines within a specified spatial domain to improve one or more measures of farm performance while satisfying geometric, operational, and environmental constraints.

A primary consideration in WFLO is the interaction between turbines through their wakes. As a wind turbine extracts energy from the incoming flow, it produces a downstream region of reduced wind speed and increased turbulence. A downstream turbine operating within this wake therefore

experiences different inflow conditions than a turbine exposed to the undisturbed free stream, which can reduce its power production.

Wake interactions are also dependent on wind direction. Turbines that are well separated perpendicular to one inflow direction may become aligned when the wind direction changes. Similarly, changes in wind speed affect both turbine operating conditions and wake behavior. Consequently, the performance of a candidate layout generally cannot be characterized by a single wind condition and must instead be considered across a representative set or distribution of environmental conditions.

WFLO is additionally subject to spatial constraints. These may include minimum separation between turbines, restrictions imposed by the available farm area, and setbacks from property or site boundaries. Environmental constraints can introduce further limitations on otherwise favorable turbine locations. Of particular relevance to this study is aeroacoustic noise, since changing turbine locations alters the distance and direction between individual turbines and surrounding acoustic receivers.

The combination of wake interactions, changing environmental conditions, and spatial constraints produces a complex optimization problem. Evaluating large numbers of candidate layouts can become particularly computationally expensive when higher-fidelity turbine and wake simulations are required. For this reason, wind-farm layout optimization can benefit from lower-cost models and optimization techniques capable of efficiently exploring a large spatial search domain.

2.2 OpenFAST and FAST.Farm

OpenFAST is an open-source wind-turbine simulation framework developed from the earlier FAST v8 (Fatigue, Aerodynamics, Structures, and Turbulence) software originally developed by NREL¹. It operates as a modular “glue-code” framework, coupling specialized models for aerodynamics, structural dynamics, control systems, hydrodynamics, and other physical processes to perform time-domain aero-hydro-servo-elastic simulations.

FAST.Farm extends OpenFAST from individual turbines to wind-farm-level simulation. Each turbine is represented by an OpenFAST instance, while FAST.Farm models the wake interactions between turbines and the surrounding wind-field dynamics. Energy extracted by an upstream turbine produces a wake characterized by reduced wind speed and increased turbulence, which can alter the inflow and resulting performance of downstream turbines.

This higher level of physical modeling allows FAST.Farm and OpenFAST to predict time-varying turbine quantities such as generator power while accounting for farm-level wake interactions. OpenFAST also includes an AeroAcoustics model within AeroDyn for predicting aerodynamic noise at specified observer locations. These capabilities make the OpenFAST and FAST.Farm framework suitable as the higher-fidelity simulation basis used in this study.

2.3 FLORIS

FLORIS (FLOw Redirection and Induction in Steady State) is an open-source, reduced-order wind-farm modeling framework developed by NREL. It uses computationally efficient engineering wake models to estimate steady-state wake interactions, turbine inflow velocities, and power production for a specified wind-farm layout and inflow condition.

Compared with time-domain simulation using FAST.Farm and OpenFAST, FLORIS can evaluate wind-farm conditions at substantially lower computational cost. This makes it suitable for repeated

¹The National Renewable Energy Laboratory (NREL) was renamed the National Laboratory of the Rockies (NLR) in 2025. Historical names are retained throughout this report when referring to work developed or published under NREL.

evaluations within an optimization algorithm, while higher-fidelity simulations can be reserved for validation.

2.4 NREL 5-MW Reference Wind Turbine

The NREL 5-MW Reference Wind Turbine is a hypothetical, standardized 5-MW baseline turbine developed by the NREL, primarily to support offshore wind-energy research and the development and comparison of wind-turbine simulation tools.

Although originally developed for offshore system research, the reference model also includes a onshore configuration. The fundamental turbine design is retained across these configurations, while the support structure is adapted for different applications. The onshore configuration uses a tower cantilevered directly to a concrete foundation, whereas offshore configurations have been developed using support structures such as monopiles, space frames, and floating platforms.

The reference turbine is a three-bladed, upwind, variable-speed, collective-pitch machine with a rated power of 5 MW, a rotor diameter of 126 m, and a hub height of 90 m. Selected properties relevant to this study are summarized in Table 1.

Table 1: Selected properties of the NREL 5-MW Reference Wind Turbine.

Property	Value
Rated power	5 MW
Rotor orientation	Upwind
Number of blades	3
Rotor diameter	126 m
Hub height	90 m
Cut-in wind speed	3 m/s
Rated wind speed	11.4 m/s
Cut-out wind speed	25 m/s
Control	Variable speed, collective pitch

The onshore configuration was selected because this study considers aeroacoustic constraints within a comparatively compact terrestrial wind farm. The optimization domain spans only 3 km, placing potential acoustic receivers at distances where turbine noise can directly influence layout selection. In contrast, utility-scale offshore wind farms may be located many kilometers from populated shoreline receptors, making community aeroacoustic constraints less directly influential on turbine placement within the farm. The onshore configuration therefore provides a more appropriate setting for investigating the tradeoff between wake-dependent power production, turbine placement, and aeroacoustic constraints.

2.5 Aeroacoustic Modeling

Wind turbines generate sound through both mechanical and aerodynamic mechanisms. For modern utility-scale turbines, aerodynamic noise produced by the interaction between the rotating blades and the surrounding airflow is an important component of the overall acoustic emission. The resulting sound depends on turbine operating conditions, blade aerodynamics, incoming flow conditions, and the position of the acoustic observer relative to the turbine.

It is important to note that due to the relative strengths of the noise outputs, OpenFAST only computes the noise produced through aerodynamic mechanisms, ignoring the noise generated

from the internal components, such as the drivetrain or gearbox. The former produces a type of noise called Amplitude-Modulated Broadband Aerodynamic Noise, which can be described as a “whooshing,” “swishing,” or “cyclic swooshing” noise that covers the whole broadband spectrum. The latter however, can be approximated as a tonal sound, which has a distinct pitch, similar to a music note.

Several aerodynamic mechanisms can contribute to wind-turbine noise, including turbulent-inflow noise and airfoil self-noise. Turbulent-inflow noise results from atmospheric turbulence interacting with the blade, whereas self-noise is associated with interactions between the blade boundary layer and the surrounding flow. A major component of this self-noise includes tip vortex noise, which is generated by turbulent pressure fluctuations at the outermost blade tip. Because these mechanisms depend on local blade and flow conditions, the resulting acoustic field can vary with both wind speed and observer direction.

OpenFAST provides aeroacoustic modeling through the AeroAcoustics functionality of AeroDyn. The model uses the aerodynamic state of the turbine to estimate sound pressure levels at specified observer locations and can represent multiple aerodynamic noise mechanisms. This allows turbine aeroacoustic behavior to be evaluated alongside the underlying aerodynamic simulation.

As mentioned before, the AeroAcoustics model is specifically concerned with aerodynamic noise generated by the turbine blades. Mechanical noise sources, such as tonal components originating from the generator or drivetrain, are not modeled by the AeroAcoustics functionality considered in this study. Because AeroAcoustics is implemented within AeroDyn, its scope is the prediction of aerodynamically generated noise rather than mechanical noise sources associated with components such as the generator or drivetrain.

For turbulent-inflow noise, AeroDyn provides formulations based on the Amiet model as well as an option incorporating the Simplified Guidati model. The specific turbulent-inflow model and AeroAcoustics configuration used in this study are described later in the methodology.

2.6 Decibels and A-Weighting

Sound levels are commonly expressed in decibels (dB), a logarithmic measure of acoustic intensity relative to a reference value. For an acoustic intensity I and reference intensity I_0 , the sound level can be expressed as

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right). \quad (1)$$

Because the decibel scale is logarithmic, sound levels from independent sources cannot be combined through ordinary arithmetic addition. Instead, the individual levels must first be converted to the linear acoustic-energy domain. For N independent sound contributions, the combined level is

$$L_{\text{total}} = 10 \log_{10} \left(\sum_{i=1}^N 10^{L_i/10} \right). \quad (2)$$

Similarly, a time-varying sequence of sound levels can be represented by an equivalent energetic mean,

$$L_{\text{eq}} = 10 \log_{10} \left(\frac{1}{T} \sum_{t=1}^T 10^{L_t/10} \right), \quad (3)$$

where T is the number of samples being averaged.

A-weighting Human hearing does not have equal sensitivity at all frequencies. A-weighting applies a frequency-dependent weighting intended to approximate the relative sensitivity of human hearing, with the resulting sound level reported in A-weighted decibels (dBA). A-weighted sound levels are commonly used when evaluating environmental and community noise because they provide a single-number measure that accounts for this frequency-dependent sensitivity.

IEC 61672-1 defines the A-weighting response using the function

$$R_A(f) = \frac{f_4^2 f^4}{(f^2 + f_1^2) \sqrt{f^2 + f_2^2} \sqrt{f^2 + f_3^2} (f^2 + f_4^2)}, \quad (4)$$

where

$$f_1 = 20.6 \text{ Hz}, \quad f_2 = 107.7 \text{ Hz}, \quad f_3 = 737.9 \text{ Hz}, \quad f_4 = 12194 \text{ Hz}. \quad (5)$$

The corresponding A-weighting correction in decibels is

$$A(f) = 20 \log_{10} [R_A(f)] + 2.00, \quad (6)$$

where the 2.00 dB normalization produces an A-weighting correction of approximately 0 dB at 1 kHz.

For an unweighted sound-pressure level $L(f)$ at frequency f , the A-weighted sound-pressure level is therefore

$$L_A(f) = L(f) + A(f). \quad (7)$$

The resulting level is expressed in A-weighted decibels (dBA).

The distinction between logarithmic and linear acoustic quantities is particularly important when considering a wind farm containing multiple turbines. The acoustic contribution from each turbine must be combined in the linear energy domain before being converted back to dBA. The same principle applies when reducing time-varying acoustic predictions to an equivalent steady-state value.

2.7 Surrogate Modeling

High-fidelity numerical simulations can provide detailed representations of complex physical systems, but their computational cost may make them impractical for applications requiring a large number of repeated evaluations. Surrogate modeling addresses this problem by constructing a computationally inexpensive approximation of a more expensive underlying model.

A surrogate model learns or approximates the relationship between a set of input variables and the corresponding outputs of the original model. For a computational model represented generally as

$$\mathbf{y} = f(\mathbf{x}), \quad (8)$$

a surrogate constructs an approximation

$$\hat{\mathbf{y}} = \hat{f}(\mathbf{x}), \quad (9)$$

where $\hat{f}$ is designed to reproduce the behavior of f at a substantially lower computational cost. The surrogate may be constructed using methods such as interpolation, regression, or machine-learning algorithms trained on data generated by the original model.

Surrogate models are particularly useful in optimization. Population-based optimization algorithms may require thousands of evaluations of candidate solutions, making direct evaluation with a high-fidelity simulation computationally expensive. A surrogate can instead be evaluated repeatedly during the optimization while the higher-fidelity model is reserved for training, verification, or final validation.

This reduction in computational cost introduces a tradeoff in accuracy. A surrogate is only an approximation of the underlying model and its reliability depends on factors such as the quality and coverage of its training data, the selected input representation, and the predictive capability of the surrogate model. Consequently, surrogate predictions should be evaluated against the higher-fidelity model they are intended to approximate.

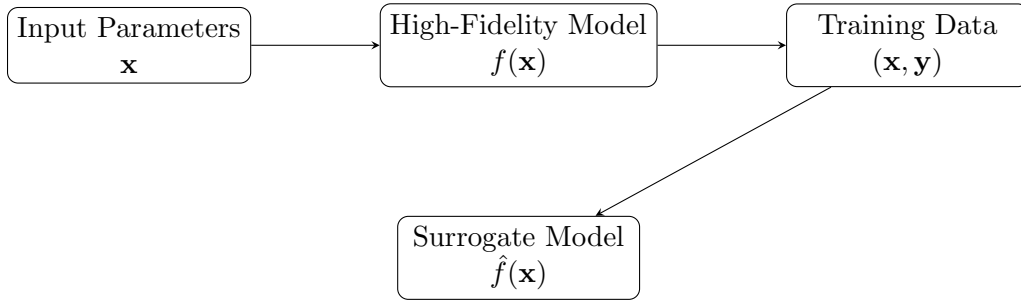


Figure 1: General concept of constructing a surrogate model from high-fidelity simulation data.

2.8 Genetic Algorithms

Genetic algorithms (GAs) are population-based optimization methods inspired by the processes of biological evolution. Rather than improving a single candidate solution, a genetic algorithm maintains a population of candidate solutions and iteratively evolves that population toward regions of higher fitness. GAs are derivative-free, meaning that they do not require an analytical gradient of the objective function. For this reason, GAs can be referred to as a type of “blind search.”

Each candidate solution is represented by a *chromosome*, whose parameters encode a possible solution to the optimization problem. A collection of chromosomes forms a *population*. Each chromosome is evaluated using a fitness function,

$$F = f(\mathbf{C}), \quad (10)$$

where $\mathbf{C}$ represents a chromosome and F is a scalar measure of the quality of that candidate solution. The definition of fitness is problem dependent and may incorporate both the optimization objective and penalties associated with constraint violations.

After fitness evaluation, genetic operators are used to construct a new generation. *Selection* favors higher-fitness chromosomes for reproduction, while *crossover* combines information from multiple parents to produce new offspring. *Mutation* introduces random changes to chromosomes, allowing new regions of the search space to be explored. GAs can also exhibit traits such as *elitism*, in which one or more of the highest-performing chromosomes are transferred directly to the next generation and *immigration*, where a bottom portion of the population is thrown out and randomly regenerated.

Through repeated evaluation and reproduction, selection tends to exploit promising regions of the search space, while crossover and mutation maintain the ability to explore alternative solutions.

The process continues until a termination criterion is reached, such as a maximum number of generations, convergence of the population, or attainment of a desired fitness.

The blindness of GAs is particularly useful for problems in which the objective function is nonlinear, discontinuous, difficult to differentiate, or evaluated through external numerical models. Their population-based nature also makes them suitable for spatial optimization problems containing multiple interacting design variables and constraints.

3 Methodology

3.1 Overall Workflow

The optimization framework developed in this study consists of two key components: FLORIS and a custom machine-learning predictor for the steady-state aeroacoustic output of each turbine at a given receiver location. These computationally efficient approximators avoid the time and computational costs associated with repeatedly running OpenFAST, making them suitable for use within the genetic algorithm. However, they are not intended to serve as a complete replacement for a higher-fidelity simulation. For this reason, a final OpenFAST/FAST.Farm validation step is included.

During the optimization stage, a genetic algorithm evaluates candidate turbine layouts using FLORIS and the machine-learning aeroacoustic surrogate model. On the first iteration, chromosomes are generated from a uniform random distribution within the constraints of the land plot. FLORIS is used to estimate wake interactions, effective turbine wind speeds at the hub, and overall farm power production for each candidate. The effective turbine wind speeds are then supplied to the aeroacoustic surrogate along with the candidate layouts to estimate the A-weighted sound level at each receiver per turbine per candidate.

The outputs of these models are incorporated into the genetic algorithm by providing information to calculate candidate fitness. The fitness function takes into account power production, as well as turbine-spacing, farm-boundary, and aeroacoustic constraints to output a single float metric. To evaluate candidates under a range of operating conditions, each generation is evaluated across a set of weighted wind-speed and wind-direction scenarios.

After each generation, chromosomes are selected based on fitness and used to produce the subsequent generation through crossover and random mutation. A portion of the lowest-performing population is replaced with newly generated random candidates to maintain population diversity, while the highest-performing candidate is retained through elitism. The specific genetic operators and optimization parameters are described in greater detail in the following sections.

Once the genetic algorithm reaches its termination criterion, currently defined as a fixed number of generations, the highest-performing candidate is evaluated using OpenFAST/FAST.Farm. The resulting higher-fidelity aeroacoustic predictions and power outputs are then used to recalculate the candidate's acoustic performance and fitness and to evaluate the agreement between the surrogate-based optimization and the OpenFAST validation.

3.2 Aeroacoustic Surrogate Model

Repeated OpenFAST aeroacoustics simulations are computationally impractical within the genetic algorithm, where many candidate layouts must be evaluated across multiple wind conditions during each generation. A machine-learning surrogate was therefore developed to approximate the steady-state A-weighted sound level produced by an individual turbine at an arbitrary receiver location.

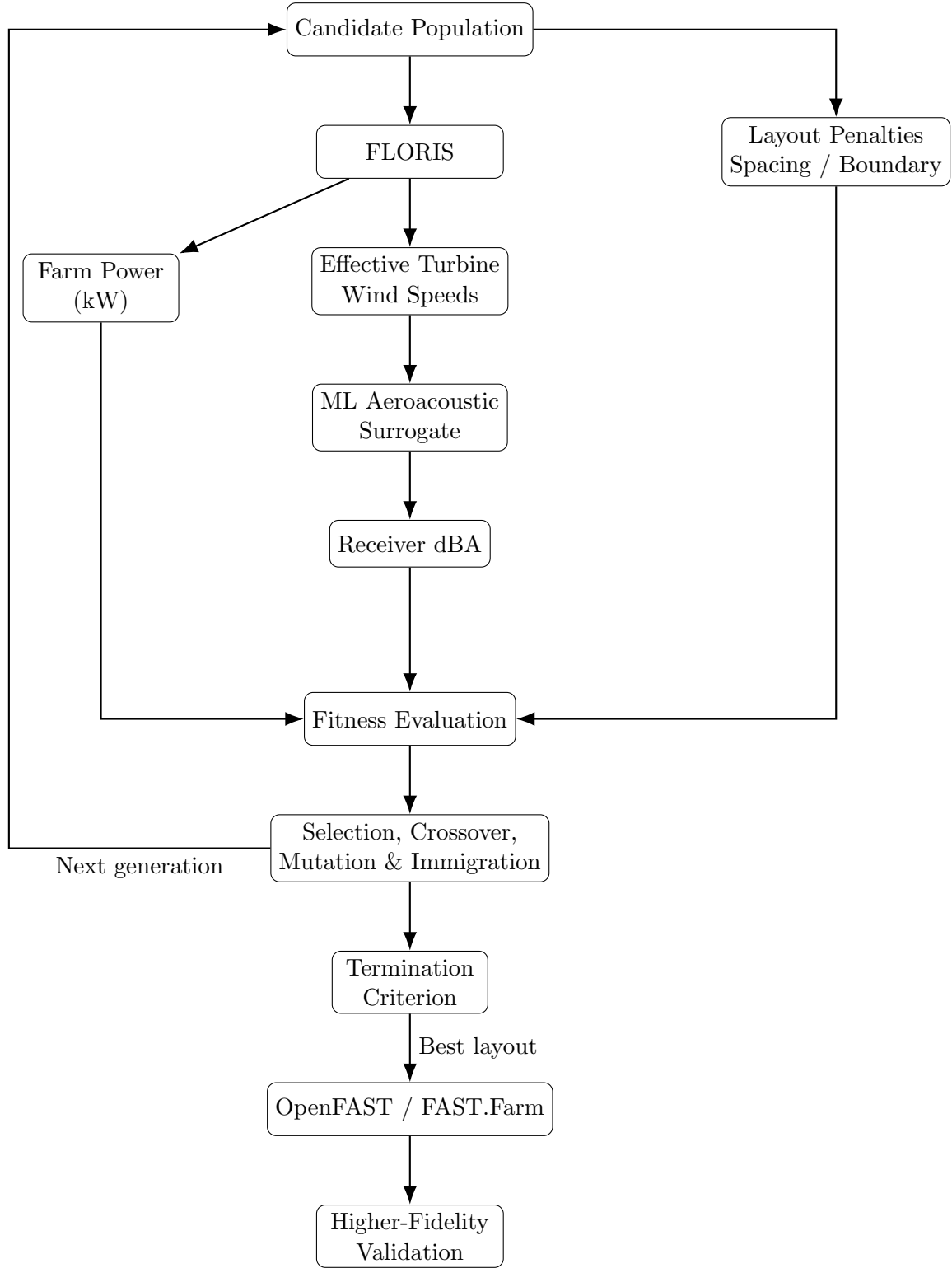


Figure 2: Overall genetic-algorithm optimization and OpenFAST/FAST.Farm validation workflow.

The surrogate was developed from single-turbine OpenFAST aeroacoustics simulations spanning a range of wind speeds and observer locations. Multiple spatial feature representations and regression algorithms were evaluated to determine the combination that most accurately reproduced the OpenFAST results. The selected model was subsequently vectorized to efficiently predict the individual turbine contributions across all turbines, receivers, and candidate layouts required during optimization.

The following sections describe the initial stages of the basic interpolator, generation of the training dataset, the feature and model selection procedure, the final surrogate implementation, and the method used to aggregate individual turbine predictions into farm-level receiver sound levels.

3.2.1 Preliminary Surrogate Designs

Before developing the machine-learning surrogate, two simpler aeroacoustic prediction approaches were investigated. The first used a simplified analytical relationship between turbine sound-power level, wind speed, and propagation distance. The second incorporated OpenFAST aeroacoustics results through interpolation over wind speed and observer azimuth. These preliminary approaches helped identify the variables and physical behavior that ultimately needed to be represented directly in the machine-learning surrogate.

Simplified Analytical Model. The initial model estimated the A-weighted sound-power level of an individual NREL 5 MW turbine as a function of incoming wind speed. A piecewise relationship was assumed in which the turbine produced no acoustic contribution below 3 m/s, increased linearly from 90 dBA to 106 dBA between 3 and 11 m/s, and remained at 106 dBA for wind speeds greater than or equal to 11 m/s, directly following from the specifications in Table 1. The assumed relationship was

$$L_w(v) = \begin{cases} 0, & v < 3, \\ 90 + 2(v - 3), & 3 \leq v < 11, \\ 106, & v \geq 11, \end{cases} \quad (11)$$

where v is the incoming wind speed in m/s and L_w is the estimated A-weighted sound-power level.

The sound-pressure level at each receiver was subsequently approximated using

$$L_p(v, d) = L_w(v) - 20 \log_{10}(d) - 8 - 0.005d, \quad (12)$$

where d is the three-dimensional turbine-to-receiver distance. This relationship was adopted as a simplified propagation approximation during early development and should not be interpreted as a complete implementation of an established acoustic propagation standard.

Although computationally inexpensive, this formulation depends only on wind speed and turbine-to-receiver distance. It therefore predicts the same sound level for receivers at equal distances regardless of their azimuth relative to the turbine and incoming wind direction. Examination of OpenFAST aeroacoustics results showed substantial directional variation, indicating that observer azimuth needed to be represented explicitly.

OpenFAST-Based Angular Interpolator. To incorporate this directional dependence, a second surrogate was constructed from single-turbine OpenFAST aeroacoustics simulations. Wind speed was varied from 3.0 to 24.5 m/s in increments of 0.5 m/s,

$$v \in \{3.0, 3.5, 4.0, \dots, 24.5\}, \quad (13)$$

resulting in 44 distinct wind-speed conditions, following from the cut-in and cut-out speeds in Table 1. For each condition, 32 acoustic observers were uniformly distributed around the turbine at a radius of 200 m.

The latter half of each aeroacoustics output time series was retained to approximate steady-state operation. An arithmetic mean of the A-weighted sound levels over this interval was then calculated for each observer. The resulting values were stored in `noise_lookup_table.json`, forming a two-dimensional lookup table parameterized by wind speed v and observer azimuth θ .

A coordinate transformation was required when coupling this dataset with FLORIS because FLORIS specifies wind direction using the meteorological “direction from which the wind originates” convention, whereas the aeroacoustic lookup requires the corresponding Cartesian source-relative direction. The conversion used in the implementation was

$$\theta_{\text{src}} = (90^\circ - \phi_{\text{met}}) \frac{\pi}{180^\circ}, \quad (14)$$

where ϕ_{met} is the meteorological wind direction and θ_{src} is the corresponding Cartesian angle in radians.

The resulting OpenFAST data were interpolated over wind speed and azimuth using a linear `RegularGridInterpolator`. For receiver locations away from the 200 m reference radius, distance dependence was still approximated analytically using logarithmic geometric spreading and the constant 0.005 dBA/m attenuation coefficient.

This approach captured the strong directional variation absent from the initial analytical model, but the remaining analytical distance correction introduced another limitation. Comparison with additional OpenFAST simulations at varying observer distances showed that a single constant attenuation coefficient did not reproduce the distance-dependent behavior consistently across all observer azimuths and wind speeds. In some conditions the correction overestimated attenuation, while in others it underestimated it.

These observations motivated the final data-driven approach. Rather than interpolating only over wind speed and azimuth while prescribing the distance dependence analytically, the subsequent surrogate was trained to learn the joint relationship between wind speed, observer distance, observer direction, and A-weighted sound level directly from OpenFAST aeroacoustics data. A larger, homogeneous dataset was therefore generated for machine-learning model development, as described in the following sections.

It should be noted that this preliminary implementation averaged decibel values directly in the logarithmic domain. The later OpenFAST validation methodology instead uses energetic averaging, as described in Section 2.6.

3.2.2 Aeroacoustic Training Data

The training data for the aeroacoustic surrogate were generated using single-turbine OpenFAST simulations. Because each turbine was simulated independently, wake interactions between turbines were not included in the training-data generation process. Wake effects are instead accounted for later in the optimization workflow using FLORIS, which provides the effective wind speed experienced by each turbine.

Two primary variables were varied when generating the aeroacoustic training data: turbine wind speed and the position of the acoustic observer relative to the turbine. The wind-speed range was the same as Set (13).

Observer locations were defined in polar coordinates using radial distance r and azimuth angle θ . Twenty radial distances were generated using logarithmic spacing between 100 and 1000 m. The reference radius of 200 m was additionally included from the previous surrogate iterations to provide a higher-resolution angular sampling at a representative observer distance. Each of the logarithmically spaced radii was evaluated at 16 azimuth angles, while the 200 m reference radius was already evaluated at 32 azimuth angles.

The resulting observer sampling therefore contained

$$(20 \times 16) + 32 = 352 \quad (15)$$

observer positions. Combining these positions with the 44 wind-speed conditions produced a total of

$$N_{\text{samples}} = 44 \times 352 = 15,488 \quad (16)$$

single-turbine aeroacoustic training samples.

Table 2 summarizes the sampling strategy.

Table 2: OpenFAST aeroacoustic training-data sampling.

Variable	Sampling	Number of Values
Wind speed v	3.0–24.5 m/s, 0.5 m/s increments	44
Radial distance r	Logarithmically spaced, 100–1000 m	20
Azimuth θ	16 angles per logarithmically spaced radius	16
Reference radius	200 m with 32 azimuth angles	32 observer positions
Total observer positions	$(20 \times 16) + 32$	352
Total training samples	44×352	15,488

Train–Test Split The complete aeroacoustic dataset was randomly divided into an 80/20 training–test split. The training set, containing 80% of the samples, was used to fit each model–feature-set combination, while the remaining 20% was held out from training and used to evaluate model performance. MAE, RMSE, and R^2 were calculated exclusively on this held-out test set to evaluate each model’s ability to generalize to unseen samples. The model did not receive any input on its performance on the test split during training. All the information and figures compiled in Section 4.1 were based on the model’s performance on the test split.

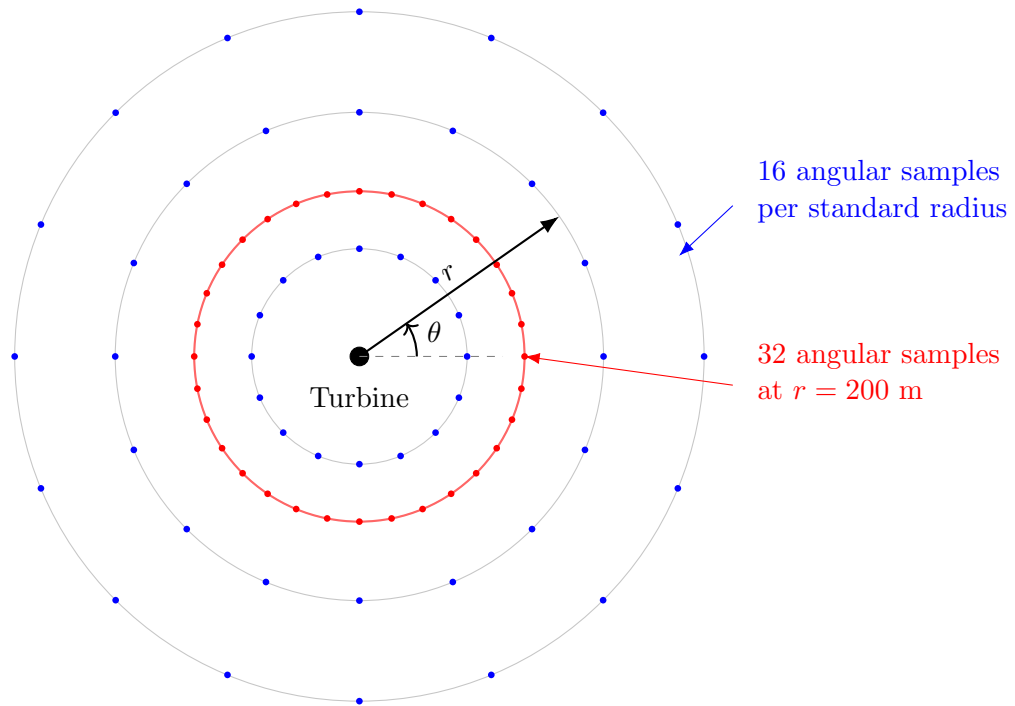
3.2.3 Candidate Feature Representations

The data generated from the OpenFAST simulations initially contained three input variables: wind speed v , relative observer coordinate X , and relative observer coordinate Y . Although the Cartesian coordinates fully describe the observer location, the aeroacoustic field exhibits a strong radial and directional structure. Several alternative feature representations were therefore evaluated to make this underlying geometry more explicit to the machine-learning models.

The relative Cartesian coordinates were first transformed into polar coordinates according to

$$r = \sqrt{X^2 + Y^2}, \quad (17)$$

and



Schematic only: blue radial spacing is logarithmic from 100–1000 m

Figure 3: Schematic representation of the observer sampling used to generate the single-turbine aeroacoustic training dataset. Standard radial distances use 16 uniformly distributed azimuth angles, while the 200 m reference radius uses a denser 32-angle sampling.

$$\theta = \text{atan2}(Y, X). \quad (18)$$

Directly using θ as a numerical feature introduces an artificial discontinuity at the angular boundary. For example, directions immediately below 2π and immediately above 0 are physically adjacent but numerically distant. To preserve the periodic nature of the observer azimuth, the angular coordinate was therefore encoded using

$$c_\theta = \cos(\theta), \quad s_\theta = \sin(\theta). \quad (19)$$

Distance was also represented using several transformations. Under geometric spreading, acoustic intensity decreases with increasing distance, while sound-pressure level is expressed on a logarithmic scale. A logarithmic transformation of the observer distance,

$$r_{\log} = \log_{10}(r), \quad (20)$$

was therefore evaluated as a candidate feature. Reciprocal distance terms were additionally considered,

$$r^{-1} = \frac{1}{r}, \quad r^{-2} = \frac{1}{r^2}, \quad (21)$$

to determine whether explicitly representing distance-decay relationships improved predictive performance.

Rather than selecting one representation a priori, five candidate feature sets were constructed and evaluated with each candidate machine-learning model. These feature sets are summarized in Table 3. The complete set of original and engineered features is retained in `training_data.csv`.

Table 3: Candidate feature representations evaluated for the aeroacoustic surrogate.

Feature Set	Input Features	Purpose
Cartesian	v, X, Y	Original spatial representation
Polar	$v, r, \cos \theta, \sin \theta$	Explicit radial and directional representation
Polar + log distance	$v, \log_{10}(r), \cos \theta, \sin \theta$	Logarithmic distance representation
Polar + raw/log distance	$v, r, \log_{10}(r), \cos \theta, \sin \theta$	Combined raw and logarithmic distance
Polar + distance decay	$v, \log_{10}(r), r^{-1}, r^{-2}, \cos \theta, \sin \theta$	Multiple representations of distance decay

The feature representation was not selected independently of the regression model. Instead, each candidate machine-learning algorithm was trained and evaluated using all five feature sets. The final combination of model architecture and feature representation was selected based on predictive performance, as described in the following section.

3.2.4 Model Training and Selection

Seven regression algorithms were evaluated for the aeroacoustic surrogate: Linear Regression, Ridge Regression, Polynomial Ridge Regression, Random Forest, Extra Trees, Histogram Gradient Boosting, and eXtreme Gradient Boosting (XGBoost). These models were selected to provide a range

of model complexities, from simple linear baselines to nonlinear ensemble and gradient-boosting methods. The normal Gradient Boosting was not considered due to the amount of rows in the training data. For $\sim 16k$ rows, XGB and HistGrad would be much more performant.

On that same note, standard Kriging and Gaussian Process Regression (GPR) would also run out of memory. The computational costs of $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space complexity would not outweigh the additional benefits of having a complete predictive probability distribution rather than a single point estimate. For around 16k rows, this would require an estimated 4 billion operations and 256 million matrix elements, which would take up ~ 2 GB of RAM for just one matrix.

Table 4 summarizes the candidate regression algorithms.

Table 4: Candidate regression algorithms evaluated for the aeroacoustic surrogate.

Model	Model Type	Role in Comparison
Linear Regression	Linear	Simple baseline model.
Ridge Regression	Regularized linear	Linear model with L_2 regularization applied.
Polynomial Ridge	Nonlinear polynomial	Higher-degree polynomial model with L_2 to avoid overfitting.
Random Forest	Tree ensemble	Nonlinear ensemble based on averaging predictions from multiple decision trees.
Extra Trees	Tree ensemble	Highly randomized tree ensemble providing an alternative to Random Forest.
Histogram Gradient Boosting	Gradient boosting	Nonlinear boosted-tree model designed for efficient regression on continuous data.
XGBoost	Gradient boosting	Alternative gradient-boosted tree implementation used to compare performance with Histogram Gradient Boosting.

Model architecture and feature representation were evaluated jointly. Each of the seven regression algorithms was trained using each of the five feature representations described in Table 3, resulting in

$$N_{\text{configurations}} = 7 \times 5 = 35 \quad (22)$$

candidate model configurations.

The dataset was divided into training and testing subsets using an 80/20 split. The same split was used for all model and feature-set combinations to provide a consistent basis for comparison.

Predictive performance was evaluated primarily using mean absolute error (MAE) and root mean squared error (RMSE), between the A-weighted sound level obtained from OpenFAST aeroacoustics, L_i , and the corresponding surrogate prediction $\hat{L}_i$. The coefficient of determination, R^2 , was also calculated as an additional measure of model fit. The prediction residual for sample i was defined as

$$e_i = y_i - \hat{y}_i, \quad (23)$$

where y_i is the OpenFAST AeroAcoustics value and $\hat{y}_i$ is the corresponding surrogate prediction. Thus, positive residuals represent underprediction by the surrogate and negative residuals represent overprediction.

The final surrogate configuration was selected based on its performance on the held-out test data. The comparative results of the 35 candidate configurations are presented in Section 4.1.

3.2.5 Final Surrogate Implementation

From our results, we can see that HistGrad with the `polar_log10_r` feature set performed the best, with the lowest MAE, RMSE, and the highest R^2 of all 35 model configurations. Thus, the machine-learning surrogate uses the input feature vector

$$\mathbf{x} = [v, \log_{10}(r), \cos(\theta), \sin(\theta)]. \quad (24)$$

The surrogate then approximates the relationship

$$\hat{L}_A = f(v, \log_{10}(r), \cos(\theta), \sin(\theta)), \quad (25)$$

where $\hat{L}_A$ is the predicted A-weighted sound level.

3.2.6 Aeroacoustic Aggregation

As mentioned in Section 2.6, sound is measured on a logarithmic scale, not a linear scale. Thus, all operations on sets of dB readings must be done after converting to a power intermediary, and the output can be converted back to dB. For this reason, independent turbine noise contributions are combined in the linear acoustic-energy domain using Equation 2. Similarly, time-varying sound levels may similarly be reduced using Equation 3.

3.3 Genetic Algorithm

A genetic algorithm was implemented to optimize the layout of the turbines within the defined farm boundaries. Each candidate was evaluated through FLORIS and the machine-learning surrogate described above. Their outputs were combined with turbine-spacing and boundary constraints to produce a single scalar fitness value used to rank chromosomes.

The optimization was performed over a fixed-size population for a specified number of generations. At each generation, candidate layouts were evaluated across the prescribed weighted wind scenarios before selection, crossover, mutation, immigration, and elitism were applied to construct the subsequent generation. The following sections describe the representation of candidate layouts, fitness calculation, environmental scenarios, and genetic operators used in the optimization.

3.3.1 Chromosome Representation and Initialization

Each chromosome represents a complete candidate wind-farm layout and is stored as an $N \times 2$ matrix, where N is the number of turbines. Each row corresponds to one turbine, while the two columns contain its Cartesian x and y coordinates:

$$\mathbf{C} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_N & y_N \end{bmatrix} \in \mathbb{R}^{N \times 2}. \quad (26)$$

The initial population is generated randomly within the rectangular farm boundary. For each turbine, the x and y coordinates are independently sampled from uniform distributions,

$$x_i \sim \mathcal{U}(X_{\min}, X_{\max}), \quad y_i \sim \mathcal{U}(Y_{\min}, Y_{\max}), \quad i = 1, \dots, N. \quad (27)$$

No minimum turbine-spacing constraint is enforced during initialization. Consequently, the initial population may contain layouts in which turbines are positioned closer together than the prescribed minimum spacing. Rather than rejecting or repairing these chromosomes, spacing violations are handled through the fitness function, which penalizes layouts in proportion to the magnitude of the violation. This allows the genetic algorithm to explore the full spatial search domain while progressively favoring layouts that satisfy the spacing requirement.

For this same reason, FLORIS may sometimes warn that two hubs may be too close to each other. This is simply a by-product of the complete random generations, but does not cause any side effects besides that specific chromosome receiving a massive hit in terms of fitness, both from power lost in wake effects and the explicit C_{boundary} penalty mentioned below.

3.3.2 Wind Scenarios

Wind-farm performance depends strongly on both wind speed and wind direction. Evaluating a candidate layout under a single wind condition could therefore favor a layout that performs well under that particular condition but poorly under other likely operating conditions. To account for this variability, each chromosome is evaluated across a discrete set of weighted wind scenarios.

Wind speed and wind direction are each represented by a set of discrete values with corresponding probability weights. The Cartesian product of these distributions forms the complete set of wind scenarios. A scenario s is represented as

$$\mathcal{S}_s = (v_s, \phi_s, w_s), \quad (28)$$

where v_s is the free-stream wind speed, ϕ_s is the meteorological wind direction, and w_s is the probability weight associated with the scenario.

If wind speed v_i has weight $w_i^{(v)}$ and wind direction ϕ_j has weight $w_j^{(\phi)}$, the joint scenario weight is calculated as

$$w_{ij} = w_i^{(v)} w_j^{(\phi)}. \quad (29)$$

Assuming the individual wind-speed and wind-direction distributions are normalized,

$$\sum_i w_i^{(v)} = 1, \quad \sum_j w_j^{(\phi)} = 1, \quad (30)$$

the resulting scenario weights also satisfy

$$\sum_{s=1}^S w_s = 1, \quad (31)$$

where S is the total number of wind scenarios.

Table 5 shows the example scenarios used in testing.

3.3.3 Fitness Function

Each chromosome is assigned a scalar fitness value that balances expected farm power production against turbine-spacing, farm-boundary, and aeroacoustic constraints. The optimization objective is expressed as

Table 5: Wind-speed and wind-direction distributions used during optimization.

Variable	Value	Weight
Wind speed	11.0 m/s	0.10
	11.5 m/s	0.80
	12.0 m/s	0.10
Wind direction	260.0°	0.20
	270.0°	0.60
	280.0°	0.20

$$F = P_{\text{expected}} - C_{\text{spacing}} - C_{\text{noise}} - C_{\text{boundary}}, \quad (32)$$

where P_{expected} is the probability-weighted farm power and the remaining terms are penalties applied to constraint violations.

The individual terms are not reduced in the same manner. Power and aeroacoustic behavior depend on the wind scenario and must therefore be evaluated across the complete wind distribution. In contrast, turbine spacing and farm-boundary violations depend only on the chromosome geometry and are consequently calculated once per layout.

Expected Farm Power. As defined in Section 3.3.2, each chromosome is evaluated under S wind scenarios with normalized weights w_s .

For a farm containing N turbines, FLORIS first provides the steady-state power $P_{i,s}$ of turbine i under scenario s . The total farm power for that scenario is

$$P_s = \sum_{i=1}^N P_{i,s}. \quad (33)$$

The expected farm power used by the fitness function is then the probability-weighted average across all scenarios,

$$P_{\text{expected}} = \sum_{s=1}^S w_s P_s. \quad (34)$$

Turbine-Spacing Penalty. Turbine spacing is evaluated directly from the chromosome geometry and is therefore independent of wind scenario. For turbines i and j , their planar separation is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \quad (35)$$

The default minimum allowable spacing is three rotor diameters,

$$d_{\min} = 3D, \quad (36)$$

where D is the turbine rotor diameter.

Only separations below this threshold are penalized. The total spacing penalty is therefore

$$C_{\text{spacing}} = k_{\text{spacing}} \sum_{i < j} \max(d_{\min} - d_{ij}, 0), \quad (37)$$

where k_{spacing} is a scaling coefficient that controls the fitness cost of each meter of spacing violation. Each turbine pair is included only once in the summation.

Farm-Boundary Penalty. A setback distance of

$$d_{\text{setback}} = 200 \text{ m} \quad (38)$$

is maintained between each turbine and the rectangular farm boundary.

Since the plot is implemented as a simple rectangle, the distances to the four boundaries for turbine i are easily represented by

$$\mathbf{b}_i = \begin{bmatrix} x_i - X_{\min} \\ X_{\max} - x_i \\ y_i - Y_{\min} \\ Y_{\max} - y_i \end{bmatrix}. \quad (39)$$

A soft penalty is applied whenever any of these distances falls below the required setback:

$$C_{\text{setback}} = k_{\text{setback}} \sum_{i=1}^N \sum_{e=1}^4 \max(d_{\text{setback}} - b_{i,e}, 0). \quad (40)$$

If a turbine lies completely outside the farm boundary, at least one $b_{i,e}$ becomes negative. These cases receive an additional large penalty consisting of a fixed term and a component proportional to the distance outside the boundary,

$$C_{\text{OOB}} = \begin{cases} 0, & \text{if all } b_{i,e} \geq 0, \\ k_{\text{flat}} + k_{\text{OOB}} \sum_{b_{i,e} < 0} |b_{i,e}|, & \text{otherwise.} \end{cases} \quad (41)$$

The complete boundary penalty is consequently

$$C_{\text{boundary}} = C_{\text{setback}} + C_{\text{OOB}}. \quad (42)$$

Aeroacoustic Penalty. Aeroacoustic behavior is scenario dependent. For every wind scenario, the surrogate predicts the total A-weighted sound level at each of the M boundary receivers. The controlling acoustic value for scenario s is the loudest receiver,

$$L_{\max,s} = \max_{m=1,\dots,M} L_{m,s}. \quad (43)$$

Given an allowable sound-level limit L_{limit} , only values above the limit contribute to the penalty:

$$C_{\text{noise},s} = k_{\text{noise}} \max(L_{\max,s} - L_{\text{limit}}, 0). \quad (44)$$

Two scenario-reduction methods are supported. In the default weighted mode, the final acoustic penalty is the probability-weighted average of the scenario-specific penalties,

$$C_{\text{noise}}^{\text{weighted}} = \sum_{s=1}^S w_s C_{\text{noise},s}. \quad (45)$$

Alternatively, a conservative worst-case mode may be used,

$$C_{\text{noise}}^{\text{worst}} = \max_{s=1,\dots,S} C_{\text{noise},s}. \quad (46)$$

The selected value of C_{noise} is then substituted into Equation 32.

Table 6: Reduction methods used by the fitness-function components.

Component	Within-Scenario Reduction	Across-Scenario Reduction
Farm power	Sum turbine powers	Probability-weighted average
Aeroacoustics	Energetically aggregate turbines, then take maximum across receivers	Weighted average of violation penalties or worst-case penalty
Turbine spacing	Sum pairwise spacing violations	None; geometry is scenario independent
Farm boundary	Sum setback and out-of-bounds violations	None; geometry is scenario independent

3.3.4 Selection and Genetic Operators

A fixed population size of 100 chromosomes is maintained throughout the optimization. After all chromosomes in a generation have been evaluated, the subsequent generation is constructed using a combination of elitism, tournament selection, crossover, mutation, and random immigration. These operators are intended to preserve high-performing solutions while maintaining sufficient population diversity to continue exploring the search space.

Population The default population size is

$$N_{\text{pop}} = 100. \quad (47)$$

The population size remains constant between generations. Each new generation consists of one preserved elite chromosome, offspring generated through selection and crossover, and newly initialized random immigrants.

Elitism The highest-fitness chromosome found during the optimization is preserved without modification and inserted directly into the subsequent generation. The current implementation therefore uses an elite count of one,

$$N_{\text{elite}} = 1. \quad (48)$$

This guarantees that the best solution discovered by the genetic algorithm cannot be lost through crossover, mutation, or random replacement.

Random Immigration To maintain population diversity, a fixed proportion of each new generation is reserved for independently generated random chromosomes. The number of immigrants is calculated from the population size and immigration rate according to

$$N_{\text{immigration}} = \lfloor r_{\text{immigration}} N_{\text{pop}} \rfloor, \quad (49)$$

where $r_{\text{immigration}}$ is the configured immigration rate. With the default values

$$N_{\text{pop}} = 100, \quad r_{\text{immigration}} = 0.40, \quad (50)$$

the algorithm introduces

$$N_{\text{immigration}} = 40 \quad (51)$$

new random chromosomes per generation.

These chromosomes are generated using the same uniform initialization procedure defined in Equation 27. The remaining population capacity, excluding the preserved elite chromosome, is filled with offspring produced through selection, crossover, and possible mutation.

Random immigration continually introduces unexplored layouts into the population, helping preserve genetic diversity and reducing the likelihood of premature convergence around a locally favorable region of the search space.

The complete population of the subsequent generation can therefore be expressed as

$$N_{\text{pop}} = N_{\text{elite}} + N_{\text{offspring}} + N_{\text{immigration}}. \quad (52)$$

Using the default parameters, the 100 chromosomes in each new generation consist of one preserved elite chromosome, 59 offspring generated through selection and crossover, and 40 randomly initialized immigrants:

$$100 = 1 + 59 + 40. \quad (53)$$

Parent Selection and Crossover The remaining positions in the new generation are populated by offspring. Parents are selected independently using tournament selection. For each parent, four chromosomes are sampled randomly from the current population,

$$N_{\text{tournament}} = 4, \quad (54)$$

and the chromosome with the highest fitness among those four is selected as the parent. This procedure is repeated independently to select a second parent.

Crossover is then performed at the individual coordinate level. Given two parent chromosomes $\mathbf{C}^{(1)}$ and $\mathbf{C}^{(2)}$, a binary mask $\mathbf{M}$ of the same $N \times 2$ dimensions is generated, with each element independently sampled as

$$M_{ij} \sim \text{Bernoulli}(0.5). \quad (55)$$

The resulting child chromosome is

$$C_{ij}^{(\text{child})} = \begin{cases} C_{ij}^{(1)}, & M_{ij} = 1, \\ C_{ij}^{(2)}, & M_{ij} = 0. \end{cases} \quad (56)$$

Consequently, each x or y coordinate has an equal probability of being inherited from either parent.

Mutation Newly generated offspring are additionally subject to mutation with a default probability of

$$p_{\text{mutation}} = 0.35. \quad (57)$$

When an offspring chromosome is selected for mutation, a normally distributed displacement is independently added to each of its coordinate values. For chromosome $\mathbf{C}$, mutation can be expressed as

$$\mathbf{C}' = \mathbf{C} + \boldsymbol{\epsilon}, \quad (58)$$

where

$$\epsilon_{ij} \sim \mathcal{N}(0, \sigma_{\text{mutation}}^2). \quad (59)$$

The default mutation spread is

$$\sigma_{\text{mutation}} = 100 \text{ m}. \quad (60)$$

Thus, mutation produces unbiased random spatial perturbations with a standard deviation of 100 m in both coordinate dimensions. Mutation is applied only to offspring produced through crossover; the elite chromosome and newly generated immigrants are not mutated.

Table 7: Default genetic-algorithm parameters.

Parameter	Symbol	Default
Population size	N_{pop}	100
Elite count	N_{elite}	1
Immigration rate	$r_{\text{immigration}}$	0.40
Tournament size	$N_{\text{tournament}}$	4
Mutation probability	p_{mutation}	0.35
Mutation spread	σ_{mutation}	100 m

3.4 OpenFAST Validation

The genetic algorithm relies on FLORIS and the machine-learning aeroacoustic surrogate to make repeated layout evaluations computationally practical. These approximations allow a large number of candidate layouts and wind scenarios to be evaluated during optimization, but the final optimized layout is additionally evaluated using FAST.Farm and OpenFAST aeroacoustics as a higher-fidelity validation step.

The highest-fitness chromosome produced by the genetic algorithm is evaluated under the same wind scenarios defined in Section 3.3.2. For each scenario, the corresponding FAST.Farm simulation is generated and executed, and the resulting aeroacoustic outputs are processed to obtain the steady-state A-weighted sound level at each receiver. These results are then used to recalculate the aeroacoustic component of the fitness function without changing the optimized turbine coordinates.

3.4.1 Wind-Direction Transformation

Rather than changing the FAST.Farm inflow direction for each wind scenario, the inflow is maintained along a fixed reference direction. The complete turbine and observer geometries are instead rotated by the corresponding wind direction before each simulation. This provides an equivalent relative orientation between the incoming wind, turbines, and acoustic observers while maintaining a consistent inflow orientation within the FAST.Farm simulations.

For a point with original coordinates (x, y) , rotation about a reference point (x_c, y_c) by an angle θ is performed using

$$\begin{bmatrix} x' - x_c \\ y' - y_c \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x - x_c \\ y - y_c \end{bmatrix}. \quad (61)$$

The same transformation is applied to both turbine and receiver coordinates, preserving their relative geometry. The rotation angle is derived from the meteorological wind direction such that each physical wind scenario is represented with the FAST.Farm inflow maintained at its reference orientation.

3.4.2 Aeroacoustic Post-Processing

For each wind scenario, the aeroacoustics output from every turbine is read and reduced to its steady-state portion. The first half of each simulation output is discarded to reduce the influence of initialization and transient behavior, leaving the latter half for acoustic validation.

Unlike the preliminary surrogate described previously, which used an arithmetic mean of the logarithmic sound levels, the validation workflow uses an energetic mean to reduce the time-varying A-weighted sound level. For a sequence of T sound levels, this is calculated using Equation 3.

Because aeroacoustics is evaluated independently for each turbine, the individual turbine contributions at a common receiver are subsequently combined in the acoustic-energy domain. For N turbines, the mean A-weighted sound levels are aggregated using Equation 2, to produce a farm-level A-weighted sound level for each receiver and wind scenario.

3.4.3 Fitness Recalculation

Following completion of the FAST.Farm simulations, the fitness of the optimized chromosome is recalculated using the higher-fidelity OpenFAST outputs. Unlike the genetic-algorithm optimization stage, in which FLORIS is used to estimate farm power and the machine-learning surrogate is used to estimate aeroacoustic levels, the validation stage obtains both farm power and aeroacoustic predictions directly from the OpenFAST simulations.

For each wind scenario s , the steady-state generator power of each turbine is extracted from the corresponding OpenFAST output file using the `GenPwr` (kW) channel. To reduce the influence of simulation initialization and transient behavior, only the latter half of the time series is retained. The steady-state power of turbine i under scenario s is calculated as

$$\overline{P}_{i,s}^{\text{OF}} = \frac{1}{T} \sum_{t=1}^T P_{i,s,t}^{\text{OF}}, \quad (62)$$

where T is the number of retained time steps in the steady-state portion of the simulation.

The total OpenFAST farm power for scenario s is then

$$P_s^{\text{OF}} = \sum_{i=1}^N \bar{P}_{i,s}^{\text{OF}}, \quad (63)$$

and the expected farm power is obtained using the same scenario probabilities employed during optimization,

$$P_{\text{expected}}^{\text{OF}} = \sum_{s=1}^S w_s P_s^{\text{OF}}. \quad (64)$$

The OpenFAST aeroacoustic outputs are processed independently as described previously to obtain the maximum farm-level receiver sound level for each scenario. The corresponding scenario noise penalties are calculated using the same sound-level limit and penalty coefficients as the genetic algorithm,

$$C_{\text{noise},s}^{\text{OF}} = k_{\text{noise}} \max(L_{\text{max},s}^{\text{OF}} - L_{\text{limit}}, 0). \quad (65)$$

The scenario-specific noise penalties are then reduced using the same weighted or worst-case method selected during optimization.

The turbine-spacing and boundary penalties are not recalculated from a different model because they depend only on the geometry of the optimized chromosome. The OpenFAST-validated fitness is therefore

$$F_{\text{validated}}^{\text{OF}} = P_{\text{expected}}^{\text{OF}} - C_{\text{spacing}} - C_{\text{boundary}} - C_{\text{noise}}^{\text{OF}}. \quad (66)$$

This provides a higher-fidelity evaluation of the chromosome selected by the genetic algorithm, since both wake-dependent power production and aeroacoustic behavior are evaluated using the OpenFAST/FAST.Farm simulation framework.

For diagnostic purposes, an additional surrogate validation basis is also calculated using the OpenFAST-derived power together with the machine-learning surrogate acoustic prediction,

$$F_{\text{surrogate,OFpower}} = P_{\text{expected}}^{\text{OF}} - C_{\text{spacing}} - C_{\text{boundary}} - C_{\text{noise}}^{\text{surrogate}}. \quad (67)$$

Because Equations 66 and 67 use the same OpenFAST-derived power and geometric penalties, their difference isolates the effect of replacing the surrogate aeroacoustic prediction with the OpenFAST aeroacoustics prediction.

4 Results

4.1 Surrogate Model Performance

The 35 model-feature-set combinations described in Section 3.2 were evaluated on the same held-out test dataset. Table 8 reports the highest-performing feature representation for each regression model, ranked by RMSE.

Histogram Gradient Boosting using the `polar_log10_r` feature representation achieved the lowest overall prediction error, with an MAE of 0.0922 dBA and an RMSE of 0.1417 dBA. The corresponding coefficient of determination was $R^2 = 99.971\%$, indicating that the model explained 99.971% of the variation present in the held-out OpenFAST AeroAcoustics data.

The other nonlinear tree-based methods also performed strongly. XGBoost and Extra Trees achieved minimum RMSE values of 0.1566 dBA and 0.1593 dBA, respectively, while the best

Table 8: Best-performing feature representation for each candidate regression model.

Model	Feature Set	MAE (dBA)	RMSE (dBA)	R ² (%)
Histogram Gradient Boosting	polar_log10_r	0.0922	0.1417	99.971
XGBoost	polar_log10_r	0.1026	0.1566	99.965
Extra Trees	polar_log10_r	0.0952	0.1593	99.963
Random Forest	polar_distance_decay	0.1718	0.2884	99.880
Polynomial Ridge	polar_distance_decay	2.0108	2.9321	87.578
Ridge Regression	polar_distance_decay	3.4931	4.8970	65.351
Linear Regression	polar_distance_decay	3.4931	4.8970	65.351

Random Forest configuration produced an RMSE of 0.2884 dBA. Adding XGBoost to the candidate models therefore did not alter the final model selection.

In contrast, the linear and polynomial models produced substantially larger errors. The best Polynomial Ridge configuration had an RMSE of 2.9321 dBA, while both Linear and Ridge Regression produced minimum RMSE values of approximately 4.897 dBA. This difference indicates that the relationship between wind speed, observer position, and the resulting aeroacoustic level could not be adequately represented by the simpler regression models considered here, as was suspected.

4.1.1 Extra Plots

The performance of the candidate model–feature-set combinations is shown in Figure 4. Histogram Gradient Boosting using the `polar_log10_r` feature representation achieved the lowest overall RMSE.

Figure 5 compares the surrogate predictions against the corresponding OpenFAST AeroAcoustics values. The predictions closely follow the 1:1 relationship across the range of sound levels, consistent with the low MAE and RMSE obtained by the selected model.

Figure 6 displays the residuals at different locations around the singular turbine. The random variation with no detectable spatial pattern implies that the surrogate successfully captures the underlying physical trends, leaving only random noise and indicating no localized systematic bias relative to the OpenFAST training data.

4.2 Genetic Algorithm Performance

The evolution of the genetic algorithm was evaluated using the saved generation-level fitness, power, aeroacoustic, and penalty metrics. These results provide insight into both the convergence behavior of the optimization and the relative influence of the different components of the fitness function.

Figure 7 shows the progression of fitness throughout the optimization. Although the mean population fitness increased steadily, the global-best fitness showed little improvement after approximately the seventh generation. This early plateau suggests that the GA may have converged prematurely and that further tuning of the genetic operators or their associated hyperparameters could improve exploration of the search space.

Figure 8 shows a similar trend in expected farm power, with the global-best value plateauing early while the population mean continued to improve. The close correspondence between the power and fitness curves suggests that power production had a strong influence on the overall fitness. It

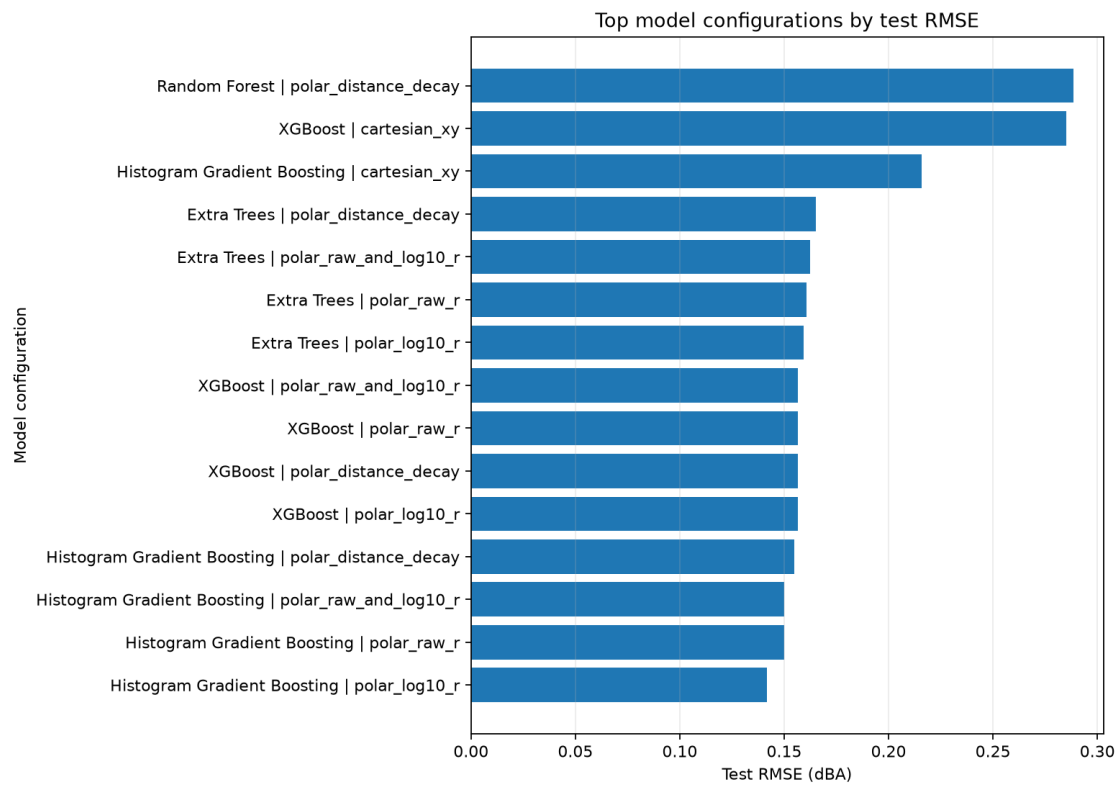


Figure 4: RMSE of the candidate model–feature-set combinations evaluated on the held-out test dataset.

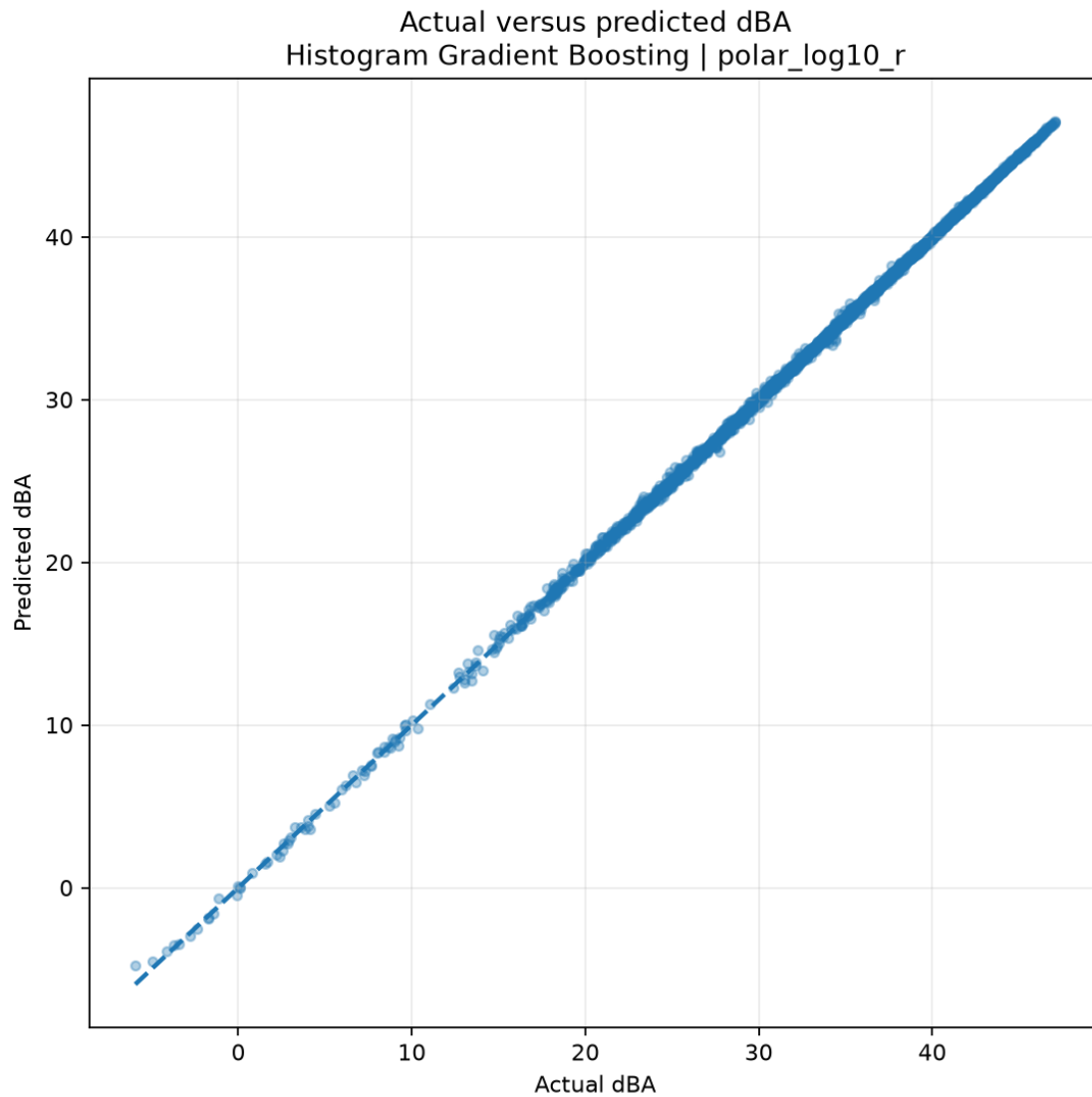


Figure 5: Predicted A-weighted sound levels from the selected surrogate model compared with the corresponding OpenFAST AeroAcoustics values in the held-out test dataset.

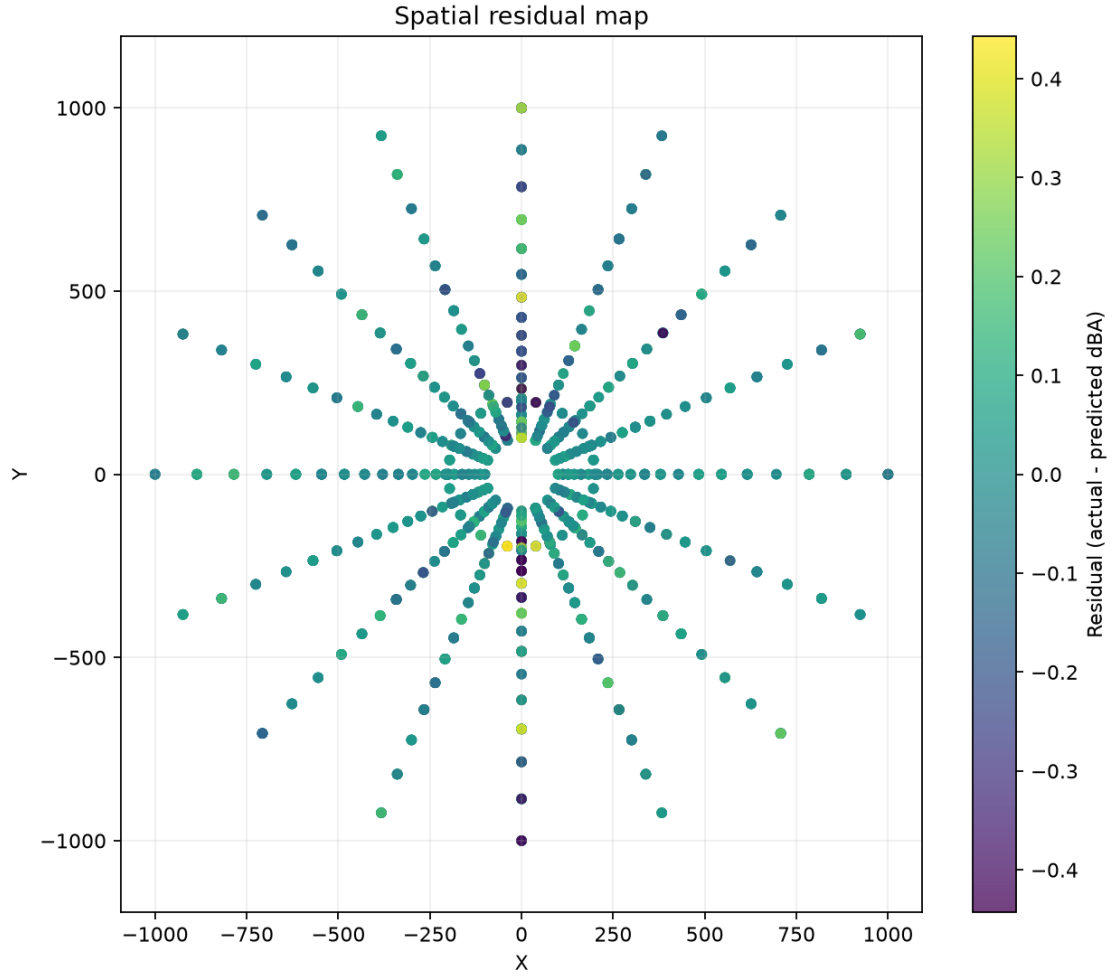


Figure 6: Spatial distribution of surrogate prediction residuals, defined as in Equation 23.

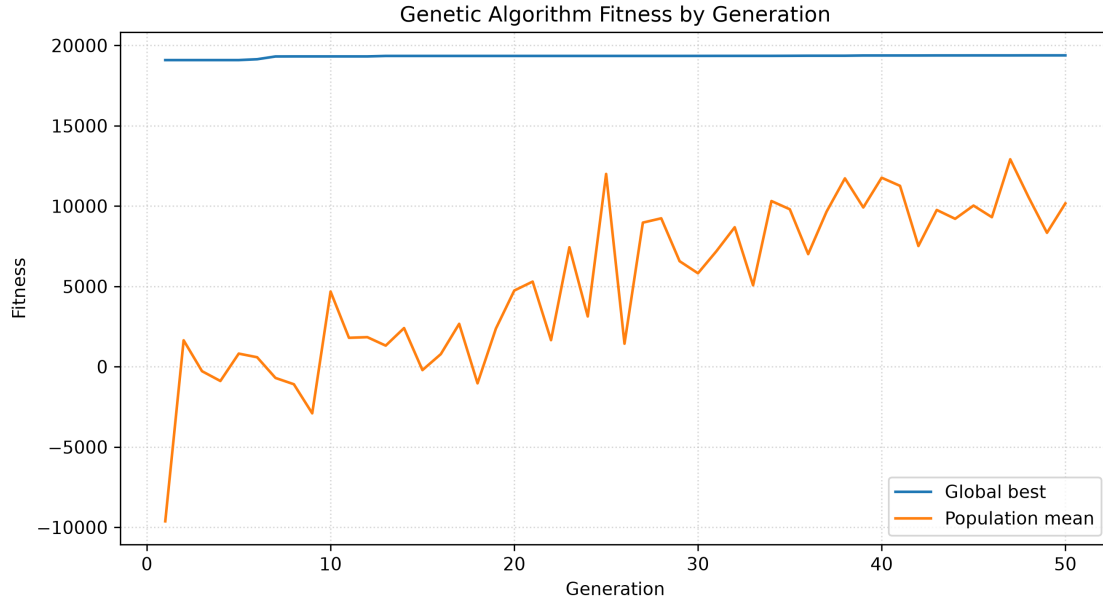


Figure 7: Genetic-algorithm fitness as a function of generation, showing the global-best fitness and mean population fitness.

may also indicate that the remaining fitness components provided relatively little differentiation among the highest-performing layouts.

Figure 9 shows the corresponding maximum receiver sound level throughout the optimization. The global-best layout remained relatively constant in its acoustic performance, although a small reduction in predicted sound level occurred during the final generations. Overall, the limited variation suggests that aeroacoustic performance changed considerably less than power and fitness during the optimization.

Figure 10 shows the mean contribution of the individual constraint penalties across the population. Spacing violations accounted for the largest penalty throughout much of the optimization, substantially exceeding the contributions from the noise and boundary constraints. Initializing chromosomes with minimum-spacing constraints already satisfied could reduce this effect and allow the GA to devote more of its search to otherwise feasible layouts.

4.3 Optimized Layout

The final optimized turbine arrangement produced by the genetic algorithm is shown in Figure 11. A distinct anisotropy is visible in the resulting geometry: the turbines are positioned substantially closer to the transverse ($\pm y$) boundaries than to the co-directional ($\pm x$) boundaries.

This behavior is consistent with the aeroacoustic directivity observed during preliminary OpenFAST AeroAcoustics testing. Across the tested wind speeds, sound levels were consistently greater in the $\pm x$ directions, approximately aligned with the wind-flow axis, than in the transverse $\pm y$ directions. The same spatial tendency was observed across repeated GA runs, in which turbine distances from the transverse boundaries were typically three to four times smaller than their distances from the co-directional boundaries. This repeated behavior indicates that the aeroacoustic constraint meaningfully influenced the spatial arrangement selected by the optimizer, rather than acting only as a scalar penalty on otherwise power-optimal layouts.

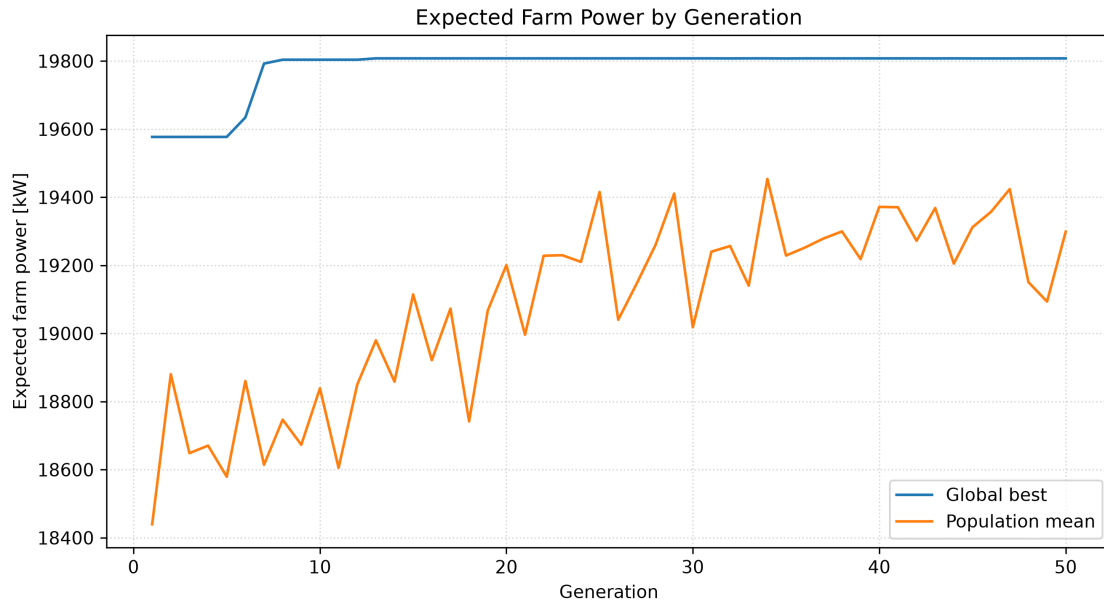


Figure 8: Expected farm power as a function of generation for the global-best chromosome and population mean.

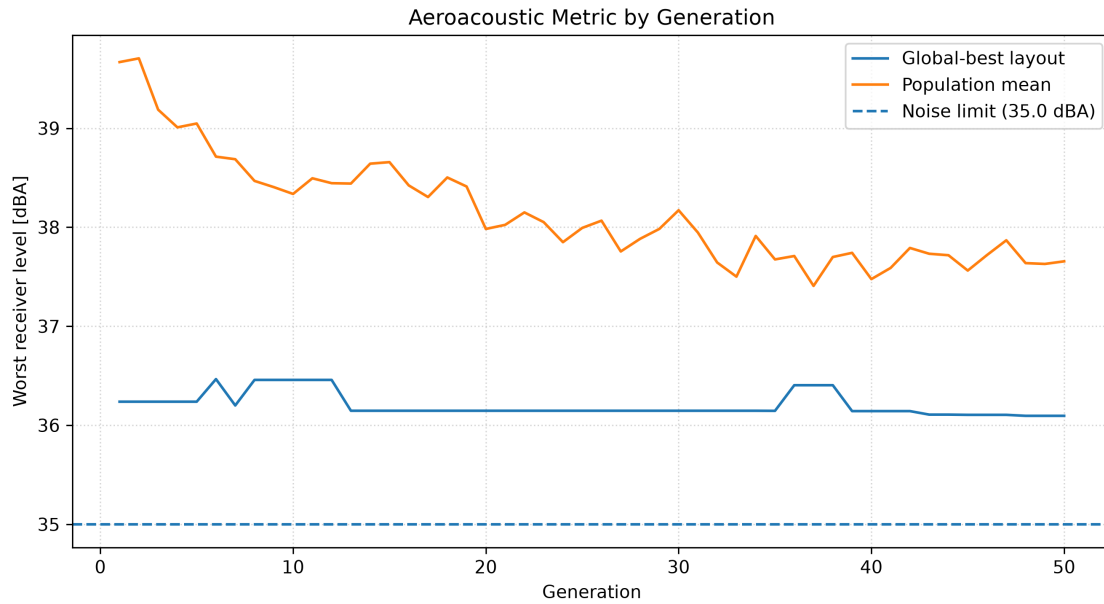


Figure 9: Maximum predicted receiver A-weighted sound level as a function of generation. The horizontal reference line indicates the configured acoustic limit.

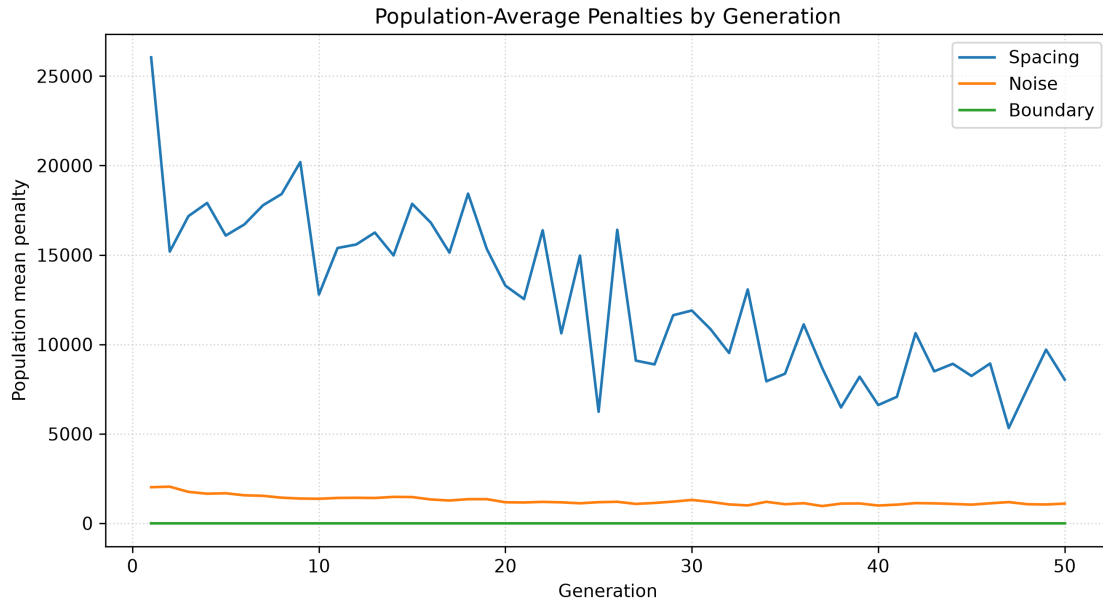


Figure 10: Mean spacing, aeroacoustic, and boundary penalties across the population as a function of generation.

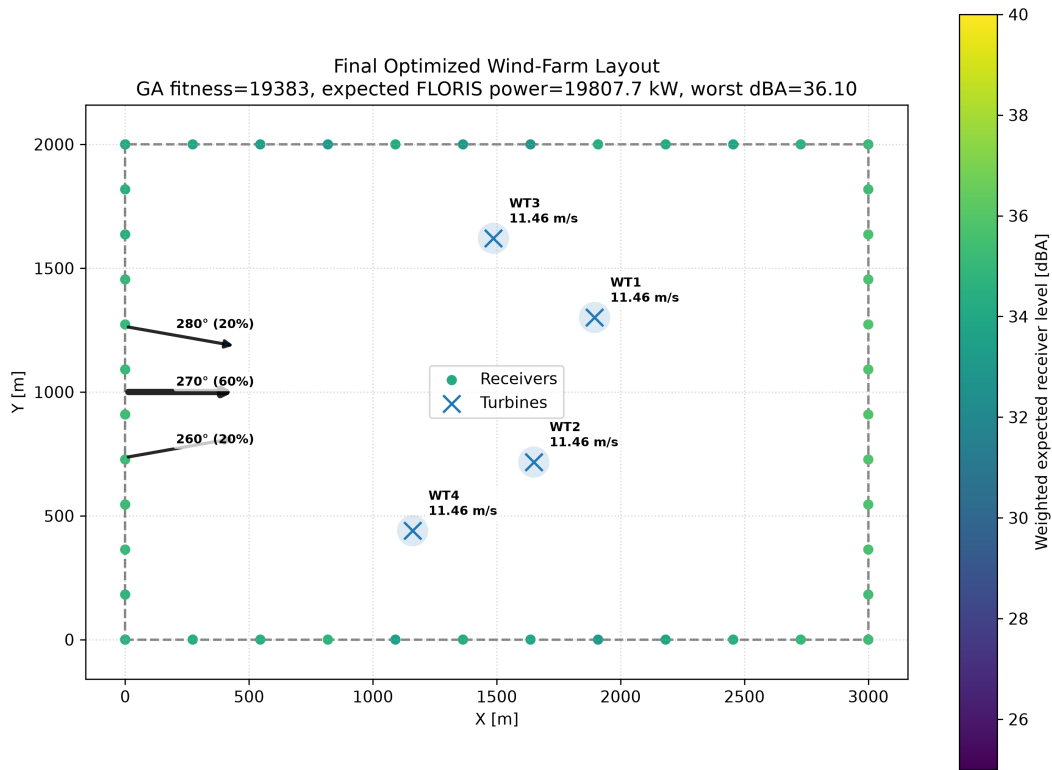


Figure 11: Final wind-farm layout identified by the genetic algorithm. Receiver shading represents the surrogate-predicted expected A-weighted sound level.

4.4 OpenFAST Validation

Although FAST.Farm also provides higher-fidelity estimates of turbine power production, a detailed comparison between FAST.Farm and FLORIS power predictions is beyond the scope of this study. The validation presented here therefore focuses specifically on the performance of the aeroacoustic surrogate when applied to turbines operating under wake-affected wind-farm conditions.

Figure 12 compares the maximum receiver sound level predicted by the aeroacoustic surrogate with the corresponding OpenFAST AeroAcoustics result for each wind scenario. The surrogate consistently predicted higher maximum sound levels than OpenFAST. In particular, the surrogate placed the maximum receiver level above the 35 dBA noise limit for every validation scenario, whereas the OpenFAST results remained at or below the limit.

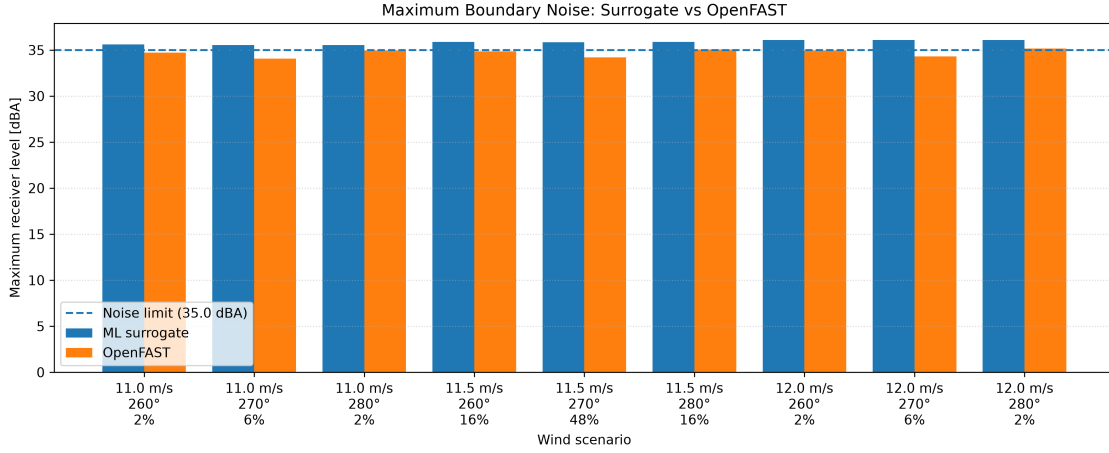


Figure 12: Maximum receiver A-weighted sound level predicted by the aeroacoustic surrogate and OpenFAST AeroAcoustics for each validation wind scenario. The dashed line represents the 35 dBA noise limit.

For the validation analysis, prediction error is defined as

$$e_{\text{validation}} = L_{\text{surrogate}} - L_{\text{OpenFAST}}. \quad (68)$$

This sign convention differs from the residual definition used during surrogate model development, where residuals were defined as observed minus predicted. Under Equation 68, positive values indicate that the surrogate predicts a higher sound level than OpenFAST, while negative values indicate surrogate underprediction.

Figure 13 shows the distribution of receiver-level validation errors defined by Equation 68. The distribution is shifted toward positive values, with a mean error of +3.024 dBA, indicating a systematic tendency for the surrogate to predict higher receiver sound levels than OpenFAST during the full wind-farm validation. This may be due to the fact that the surrogate model was not trained on waked wind, but rather steady wind, while FAST.Farm gives OpenFAST the full waked wind for use in AeroDyn and AeroAcoustics.

5 Discussion

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus. Aenean ac enim. In luctus. Phasellus eu quam vitae turpis viverra pellentesque. Duis feugiat felis ut enim.

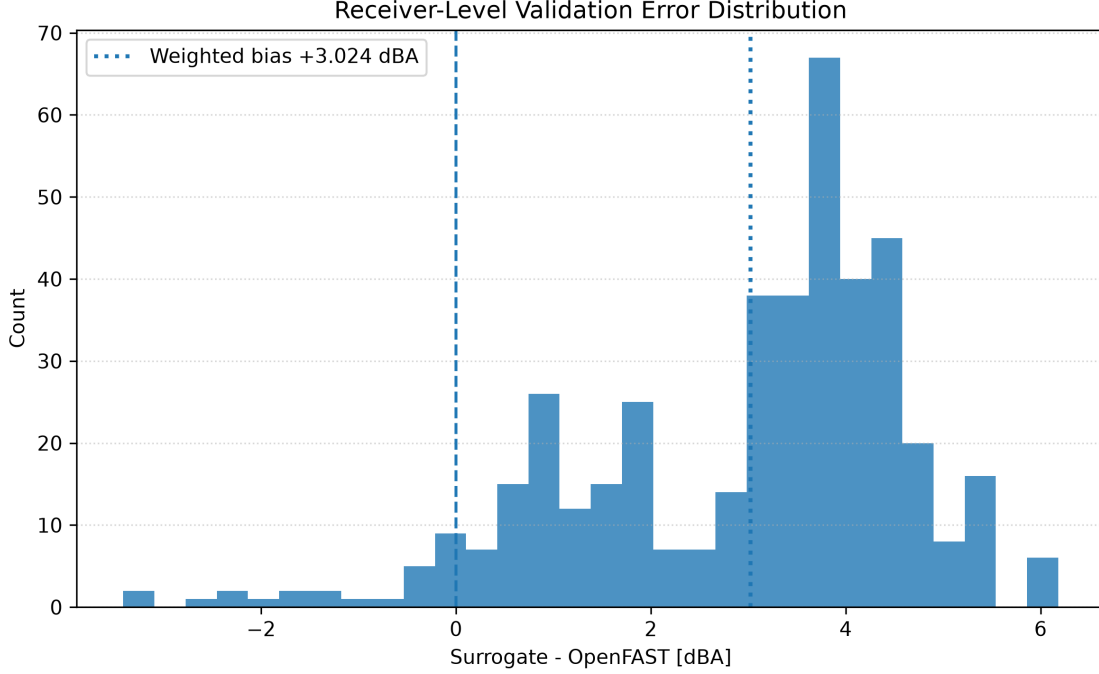


Figure 13: Distribution of receiver-level validation errors, defined as $L_{\text{surrogate}} - L_{\text{OpenFAST}}$. Positive values indicate surrogate overprediction and negative values indicate surrogate underprediction.

Phasellus pharetra, sem id porttitor sodales, magna nunc aliquet nibh, nec blandit nisl mauris at pede. Suspendisse risus risus, lobortis eget, semper at, imperdiet sit amet, quam. Quisque scelerisque dapibus nibh. Nam enim. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ut metus. Ut metus justo, auctor at, ultrices eu, sagittis ut, purus. Aliquam aliquam.

Etiam pede massa, dapibus vitae, rhoncus in, placerat posuere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor. Phasellus aliquet odio ac lacus tempor faucibus. Praesent sed sem. Praesent iaculis. Cras rhoncus tellus sed justo ullamcorper sagittis. Donec quis orci. Sed ut tortor quis tellus euismod tincidunt. Suspendisse congue nisl eu elit. Aliquam tortor diam, tempus id, tristique eget, sodales vel, nulla. Praesent tellus mi, condimentum sed, viverra at, consectetur quis, lectus. In auctor vehicula orci. Sed pede sapien, euismod in, suscipit in, pharetra placerat, metus. Vivamus commodo dui non odio. Donec et felis.

6 Limitations

Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci. Pellentesque mattis lacinia elit. Fusce molestie velit in ligula. Nullam et orci vitae nibh vulputate auctor. Aliquam eget purus. Nulla auctor wisi sed ipsum. Morbi porttitor tellus ac enim. Fusce ornare. Proin ipsum enim, tincidunt in, ornare venenatis, molestie a, augue. Donec vel pede in lacus sagittis porta. Sed hendrerit ipsum quis nisl. Suspendisse quis massa ac nibh pretium cursus. Sed sodales. Nam eu neque quis pede dignissim ornare. Maecenas eu purus ac urna tincidunt congue.

Example limitations could be presented as:

- OpenFAST validation currently uses deterministic uniform inflow.

- The aeroacoustic surrogate is valid only within the domain represented by its training data.
- The Genetic Algorithm does not guarantee identification of the global optimum.
- Additional turbulence conditions would require further high-fidelity simulations.

7 Future Work

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis. Vivamus viverra purus vel magna. Donec in justo sed odio malesuada dapibus. Nunc ultrices aliquam nunc. Vivamus facilisis pellentesque velit. Nulla nunc velit, vulputate dapibus, vulputate id, mattis ac, justo. Nam mattis elit dapibus purus. Quisque enim risus, congue non, elementum ut, mattis quis, sem. Quisque elit.

8 Conclusion

Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem. Donec ut ante quis dolor bibendum condimentum. Etiam egestas tortor vitae lacus. Praesent cursus. Mauris bibendum pede at elit. Morbi et felis a lectus interdum facilisis. Sed suscipit gravida turpis. Nulla at lectus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Praesent nonummy luctus nibh. Proin turpis nunc, congue eu, egestas ut, fringilla at, tellus. In hac habitasse platea dictumst.

References

- [1] Author, *Example OpenFAST Reference*, Publisher or Journal, Year.
- [2] Author, *Example FLORIS Reference*, Publisher or Journal, Year.

A Optimization Parameters

Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum. Suspendisse condimentum, tortor at egestas posuere, neque metus tempor orci, et tincidunt urna nunc a purus. Sed facilisis blandit tellus. Nunc risus sem, suscipit nec, eleifend quis, cursus quis, libero. Curabitur et dolor. Sed vitae sem. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Maecenas ante. Duis ullamcorper enim. Donec tristique enim eu leo. Nullam molestie elit eu dolor. Nullam bibendum, turpis vitae tristique gravida, quam sapien tempor lectus, quis pretium tellus purus ac quam. Nulla facilisi.

B Additional Results

Duis aliquet dui in est. Donec eget est. Nunc lectus odio, varius at, fermentum in, accumsan non, enim. Aliquam erat volutpat. Proin sit amet nulla ut eros consectetur cursus. Phasellus dapibus aliquam justo. Nunc laoreet. Donec consequat placerat magna. Duis pretium tincidunt justo. Sed sollicitudin vestibulum quam. Nam quis ligula. Vivamus at metus. Etiam imperdiet imperdiet pede. Aenean turpis. Fusce augue velit, scelerisque sollicitudin, dictum vitae, tempor et, pede. Donec wisi sapien, feugiat in, fermentum ut, sollicitudin adipiscing, metus.